

Education

Nov 2021 – **International Institute of Information Technology Hyderabad.**

Present B.TECH IN COMPUTER SCIENCE AND MS (BY RESEARCH) IN COMPUTATIONAL LINGUISTICS

- **CGPA:** 8.94/10.00 (Total); 9.28/10.00 (MS)
- **Dean's Academic List:** 2021–22 M (Top 5%), 2022–23 M (Top 10%), 2023–24 S (Top 15%)
- **Dean's Research List:** 2024–2025 (Top 5%)

Publications

- 2025 **PrivacyBench: A Conversational Benchmark for Evaluating Privacy in Personalized AI.**
S. Mukhopadhyay, S. Reddy, S. Muthukumar, J. An, P. Kumaraguru
WCCI (IJCNN) 2026, Nominated for Best Paper 📄 Paper
- 2025 **InterChart: Benchmarking Visual Reasoning Across Decomposed Charts.**
A. Iyengar*, **S. Mukhopadhyay***, A. Qidwai*, S. Singh, D. Roth, V. Gupta
AACL 2025 🌐 Website
- 2025 **MapIQ: Evaluating Multimodal Large Language Models for Map QA.**
V. Srivastava, F. Lei, **S. Mukhopadhyay**, V. Gupta, R. Maciejewski
COLM 2025 📄 Paper
- 2025 **PRAISE: Enhancing Product Descriptions with LLM-Driven Structured Insights.**
A. Qidwai*, **S. Mukhopadhyay***, P. Khatiwada*, D. Roth, V. Gupta
ACL 2025 (System Demonstrations) 🌐 Website
- 2025 **MAPWise: Evaluating Vision-Language Models for Advanced Map Queries.**
S. Mukhopadhyay, A. Rajgaria, P. Khatiwada, M. Shrivastava, D. Roth, V. Gupta
NAACL 2025 (Oral), Nominated for Outstanding Paper 🌐 Website
- 2024 **Unraveling the Truth: Do VLMs really Understand Charts?.**
S. Mukhopadhyay*, A. Qidwai*, A. Garimella, P. Ramu, V. Gupta, D. Roth
EMNLP 2024 🌐 Website

Research Experience

Feb 2026 – **Supervised Program for Alignment Research (SPAR).**

Present Advisor: Niels Warncke (Center on Long-Term Risk), Chad DeChant (New York University)

- Investigating how episodic memory and long-horizon reasoning affects safety and alignment in AI Agents.
- Understanding the effect of localized finetuning on generalization in LLMs.

Jan 2025 – **Precog Research Group, IIIT Hyderabad.**

Present Advisor: Prof. Ponnurangam Kumaraguru

Collaborators: Prof. Saurav Prakash (IIT-M), Prof. Sheetal Kalyani (IIT-M), Prof. Jisun An (Indiana University)

- Investigating fairness constraints in Federated Learning (FL) setups to mitigate algorithmic bias across heterogeneous clients.
- Optimizing the training of Binarized Neural Networks (BNNs) to enable efficient learning and adaptation on resource-constrained devices.
- Evaluating privacy risks in personalized agents to detect secret leakage and ensure alignment with Contextual Integrity frameworks.

- Aug 2023 – **Language Technologies Research Center (LTRC), IIIT Hyderabad.**
 Present Advisor: Prof. Manish Shrivastava *(MS Thesis Advisor)*
- Explored lateral model compression strategies to test the limits of parameter reduction in large-scale architectures.
 - Conducted comparative analysis of diverse Large Language Models (LLMs) to map representational similarities.
 - **Current Research:**
 - Investigating architectures for the addition of an effective, continually learnable memory layer in LLMs.
 - Systematically diagnosing failure modes in agents during tasks requiring long-horizon planning and backtracking.
- May 2025 – **Whitebox AI Safety Fellowship.**
 June 2025 Mentor: Kyle Reynoso
- Conducted mechanistic interpretability research to diagnose the phenomenon of emergent misalignment in Large Language Models (LLMs) using cross-encoders.
 - Analysed other diverse tasks that might lead to misaligned behaviors in models.
 - Project was ranked as Top 3 in the cohort.
- Jan 2024 – **Cognitive Computation Group (UPenn) / CORAL Lab (ASU).**
 Aug 2025 Advisors: Prof. Dan Roth (UPenn), Prof. Vivek Gupta (ASU)
 Collaborators: Dr. Aparna Garimella (Adobe), Pritika Ramu (Adobe), Prof. Ross Maciejewski (ASU)
- Investigated the robustness and consistency of Vision-Language Models (VLMs) in interpreting complex visual information, specifically charts, graphs, and geospatial maps.
 - Benchmarked the robustness of VLM reasoning capabilities on different kinds of charts, including multiple charts, identifying failure points.
 - Developed "MAPWise" and "MapIQ" benchmarks to evaluate multimodal models on advanced geospatial queries, counterfactual reasoning, and map-reading literacy.
- Jan 2024 – **Software Engineering Research Center (SERC), IIIT Hyderabad.**
 May 2024 Advisor: Prof. Karthik Vaidhyanathan
- Investigated automated code summarization techniques to extract functional natural language descriptions from code snippets, benchmarking on the APPS dataset.
 - Explored the integration of Abstract Syntax Trees (ASTs) to enrich model inputs with hierarchical structural context, aiming to enhance the semantic accuracy of generated summaries.

Industry Experience

- Aug 2025 – **Enterpret.**
 Apr 2026 Role: Machine Learning Research Intern
- Created a semantic cache to cut down costs of our retrieval pipeline by upto 30%.
 - Built evaluations to understand the limitations of our agentic system, and then fixes to improve on those areas.
 - Helped cut down the costs of the retrieval pipeline by upto 50% while also focusing on improving recall.
 - Created a pipeline for assigning more contextual names for taxonomy nodes
- May 2025 – **Google.**
 Aug 2025 Role: Software Engineering Intern Team: Google Cloud Storage (GCS) Orion
- Created a topology visualization tool for GCS Orion, enabling engineers to isolate and debug critical network issues across 100,000+ nodes.
 - Received Pre-Placement Offer (PPO) for Full-Time Software Engineer role.
- May 2024 – **Google.**
 July 2024 Role: Software Engineering Intern Team: Android PDF
- Designed and implemented the core image selection pipeline for the Android PDF viewer.
 - Received Return Internship Offer for Summer 2025 based on exceptional performance.
- May 2023 – **Google.**
 July 2023 Role: STEP Intern Team: Google Cloud Storage (GCS) Metering
- Built integrity checks for a high-throughput pipeline processing petabytes of Cloud usage data, ensuring billing accuracy for enterprise customers.
 - Converted to standard Software Engineering Intern (Return Offer) for Summer 2024.

Honors and Awards

- **India AI Fellowship (2025)**: Fellowship and aid to conduct research on LLM Similarity.
- **Amazon ML Challenge (2024)**: Secured **13th Rank** nationally among ~75,000 participants (Top 0.02%).
- **Google Generation Scholarship (2024)**: Awarded for academic excellence and leadership in technology.
- **JPMorgan Chase Quant Research Mentorship (2024)**: Selected for the specialized quantitative research track.
- **D. E. Shaw Ascend Educare (2022)**: Selected as a Fellow for the prestigious mentorship cohort as one of the youngest participants.

Teaching and Leadership

Teaching Assistantships

- Spring 2025, **Head Teaching Assistant**, *Introduction to Natural Language Processing (NLP)*.
Spring 2024 Instructor: Prof. Manish Shrivastava
- Monsoon **Teaching Assistant**, *Language and Society*.
2024 Instructor: Prof. Aditi Mukherjee
- Monsoon **Teaching Assistant**, *Algorithm Analysis & Design*.
2023 Instructor: Prof. Suryajith Chillara

Volunteership

- July 2023 **Team Leader & Mentor**, *International Linguistics Olympiad (IOL)*, Bansko, Bulgaria.
Selected to lead and mentor the Indian National Team at the global finals.
 - Trained students on advanced linguistic puzzle-solving techniques during the training camp.
 - Helped manage team strategy and logistics during the competition in Bulgaria.
- May 2023 – **Lead Student Coordinator**, *Panini Linguistics Olympiad (PLO)*, India.
Dec 2024
 - Orchestrated the national-level selection process for the Indian team, managing logistics for thousands of participants and coordinating with regional centers.
 - Helped conduct the Asia Pacific Linguistics Olympiad in India to select the Indian team.
- Mar 2023 – **Club Coordinator**, *Open Source Developers' Group (OSDG)*, IIIT Hyderabad.
Feb 2024
 - Organized an intra-college beginner hackathon attracting over 150 participants from the first and second years.
 - Conducted technical workshops on Linux systems, Open Source contributions, and Google Summer of Code (GSoC) mentorship.